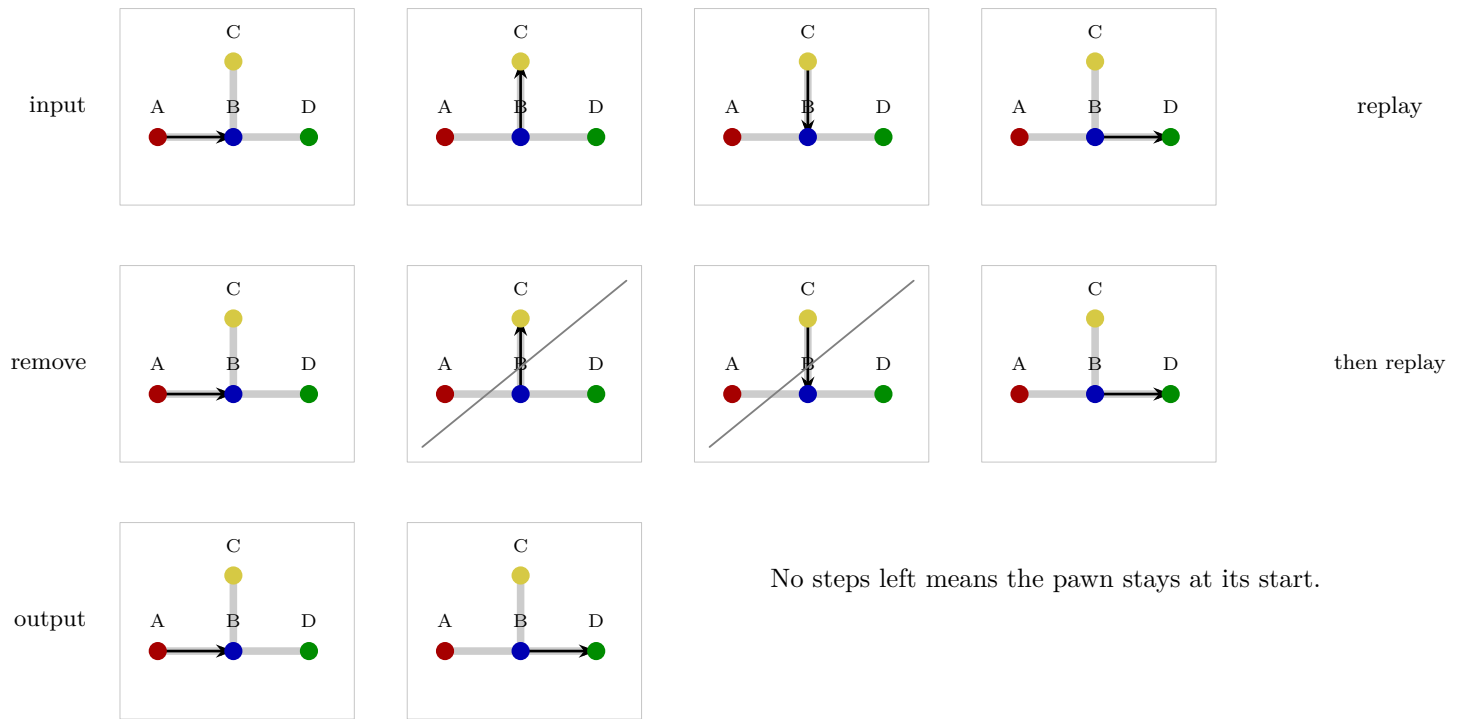


Move a pawn on the roads. Keep each journey as ordered step tiles; an adult may record. Remove only two adjacent steps along the same road in opposite directions. Replay the other steps in order. The map, start, and finish stay fixed.

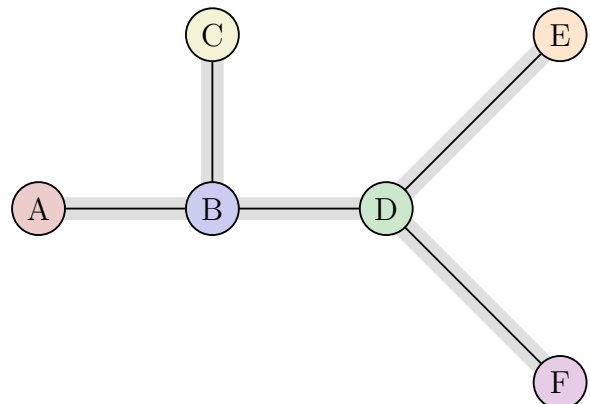


Problem 1: Shorten each recorded trip until no move is possible. Make a different seven-step trip from C to F that finishes with the same route as the last one.

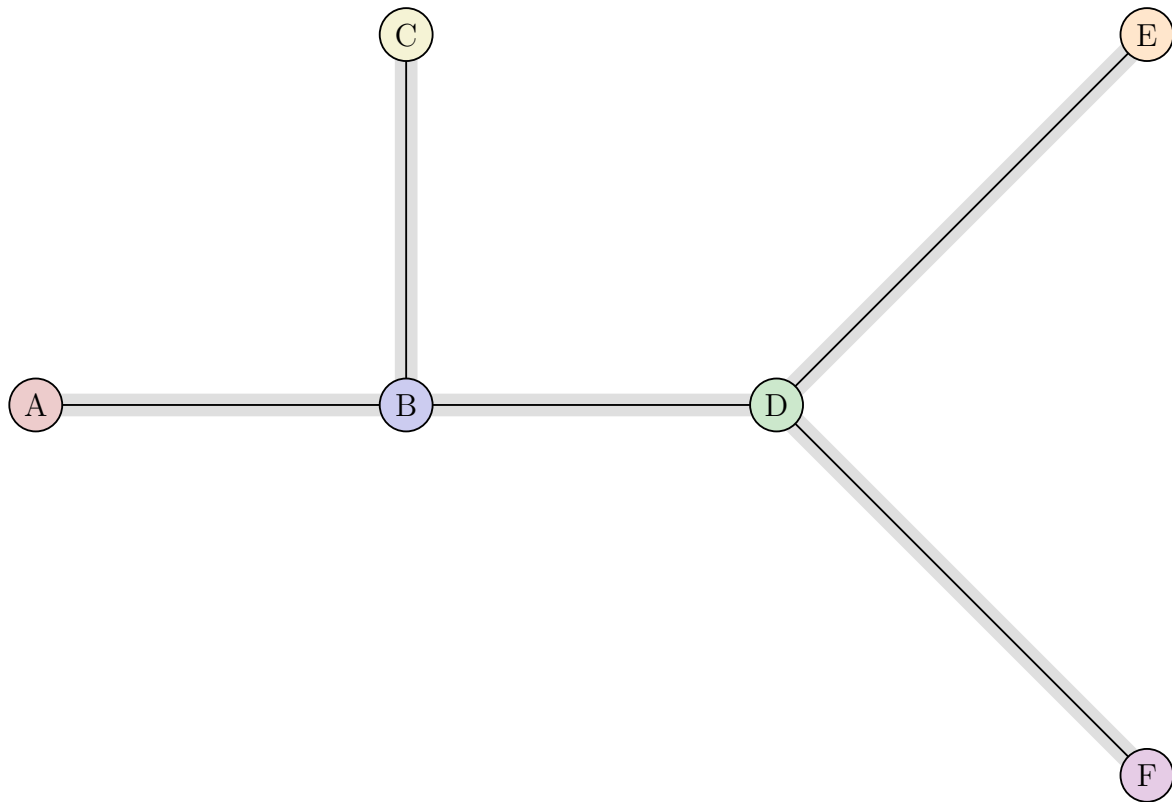
$A \rightarrow B \rightarrow A \rightarrow B \rightarrow D \rightarrow F \rightarrow D \rightarrow E \rightarrow$

$A \rightarrow B \rightarrow D \rightarrow F \rightarrow D \rightarrow E \rightarrow$

$C \rightarrow B \rightarrow C \rightarrow B \rightarrow D \rightarrow F \rightarrow$

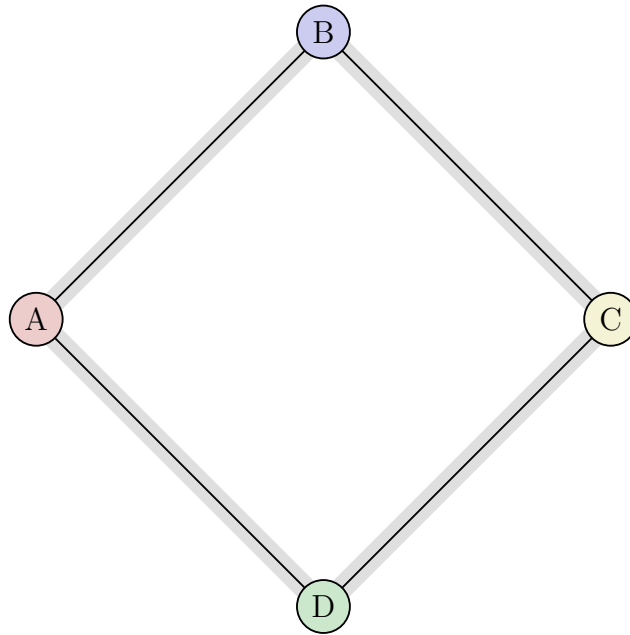


Problem 2: On this map, can a trip from A back to A survive after every immediate reversal is removed? Explain why your answer holds for trips of any length.



Problem 3: Can two trips from C to F finish with different routes after every immediate reversal is removed? Explain using the roads.

Problem 4: Shorten each recorded trip in different orders. Can the choice of first reversal ever change the final route on any road map? Explain, or give a counterexample.

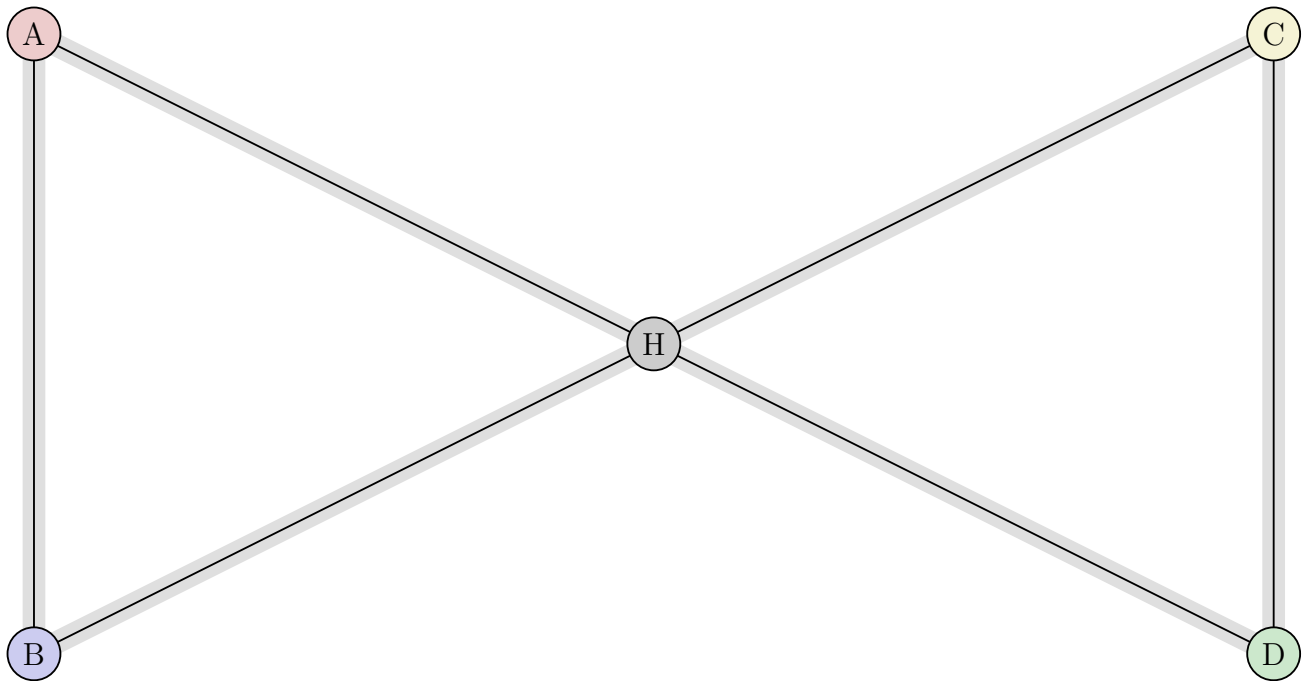


$A \rightarrow B \rightarrow A \rightarrow B \rightarrow C \rightarrow B \rightarrow A \rightarrow D \rightarrow C \rightarrow D \rightarrow A \rightarrow$ _____

$A \rightarrow B \rightarrow A \rightarrow D \rightarrow C \rightarrow B \rightarrow C \rightarrow$ _____

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow A \rightarrow B \rightarrow C \rightarrow B \rightarrow A \rightarrow$ _____

Problem 5: From H, go around the left ring through A, then the right ring through C, then the left ring backward, then the right ring backward. Can the whole trip disappear by removing immediate reversals?



Problem 6: This trip goes around each ring once each way. Does that information alone tell whether a trip can disappear? Make another trip with those same counts and a different result.